Computer Science & IT



Data Structure & Programming

Stack

Lecture No. 04



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Infix to prefix Postfix

Using stack (HPU, LPoPu)

prefix (Reverse, postfix, Reverse)

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

postfix and prefix evaluation using stack.

practice problem

Slide



Topic : Evaluation of Postfix Expression

$a+(b*c)/d-e$ Infix

To evaluate multiple Scanning required to get precedence and Associativity..

Prefix : Polish Notation

postfix : Reverse Polish Notation

polish & Reverse Polish Single scan can evaluate them Using stack

Slide



Topic : Evaluation of Postfix Expression

Conversion - stack (operator stack)

Evaluation : operand stack



Topic : Evaluation of Postfix Expression

Single digit

6 2 3 + − 3 8 2 / + * 2 ↑ 3 +

6	2	3	+ ←	−	3	8	2	/	+	*	2	↑	3	+
							2							
		3				8	8	8/2 = 4						
	2	2	2+3 = 5		3	3	3	3	3+4=7		2		3	
6	6	6	6	6-5 = 1	1	1	1	1	1	1*7=7	7	7↑2 = 49	49	49+3 = 52

Ans = 52

Stack operand Numeric value 2 variable

operand$_2$ = pop() 3 ← second pop

operand$_1$ = pop() 2

2+3

Slide



Topic : Evaluation of Postfix Expression

Single digit Binay operator

23 * 422 * / + 35 + -

2 3 * 4 2 2 * / + 3 5 + −

					2								
				2	2	2*2= 4				5			
	3		4	4	4	4	4/4 = 1		3	3	3+5 =8		
2	2	2*3 =6	6	6	6	6	6	6+1 =7	7	7	7	7-8 =-1	

−1

pop

Slide



Topic : Evaluation of Postfix Expression

opndstk = operand stack

```
while (end of Input) {
    Symb = the next Input symbol;
    if (symb is operand)
            push symb in opndstk
    else {
            operand2 = pop(opndstk);
            operand1 = pop(opndstk)
            value = operand1 symb operand2;
            push (opndstk, value);
    }
} return pop(opndstk);
```

Slide



Topic : Question

2007

Singledigit , Binary

#Q. The following postfix expression with single digit operands is evaluated using a stack:

823^ / 23 * + 51 * −

Note that ^ is the exponentiation operator. The top two elements of the stack after the first * is evaluated are:

(A) 6, 1

(B) 5, 7

(C) 3, 2

(D) 1, 5

8	2	3	^	/	2	3	*
		3				3	
	2	2	$2^3 = 8$		2	2	$2*3 = 6$
8	8	8	8	$8/8 = 1$	1	1	1

Slide

Topic : Question

Consider two binary operators '↑' and '↓' with the precedence of operator ↓ being lower than that of the operator ↑. Operator ↑ is right associative while operator ↓, is left asociative.

↑ is exponentiation operator, $a \uparrow b = a^b$

↓ is logarithmic operator $a \downarrow b = \log_b a$

The value of the expression $(65536 \downarrow 2 \uparrow 4 \uparrow 2 \downarrow 2)$ is _____.

$a \uparrow b = \boxed{a^b}$

$a \downarrow b = \log_b a$

$65536 \downarrow (2 \uparrow (4 \uparrow 2)) \downarrow 2$

$(65536 \downarrow (2 \uparrow 16)) \downarrow 2$

$(65536 \downarrow 65536) \downarrow 2$

$\log_{65536} 65536$

$(1 \downarrow 2) = \log_2 1 = 0$

$a \quad b$

Slide



Topic : Question

Q Which of the following expressions evaluates to the largest number?

(A) The postfix expression 2 3 + 5 * 7 – → 18

(B) The prefix expression + * – 2 3 5 7 → 5 * (–2) = –10

(C) The infix expression (2 + 3) * (5 - 7)

(D) The infix expression 2 + (3 * 5) – 7 → 2 + 15 – 7 = 10

23+ 5*7-

55* 7-

257-

25-7 = 18

+*-2357

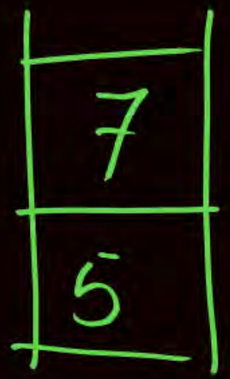
2				
3	2	3		
5	5	-1*5 = -5		
7	7	7	-5-7 = -12	

Slide

$+75 \Leftarrow 7+5 \rightarrow 75+$

$operand_1 + operand_2$





$7 \leftarrow operand_1$

$5 \leftarrow operand_2$

$+*-2357$

$+*(2-3)57 = ((2-3)*5)+7$

$+((2-3)*5)7 \quad -517 = 2$

for prefix evaluation

Scan from Right to left

1 top = $operand_1$

2 top-1 = $operand_2$



Topic : Question

Q Which of the following expressions evaluates to the largest number?

(A) The postfix expression 2 3 + 5 * 7 –

(B) The prefix expression + * – 2 3 5 7

(C) The infix expression (2 + 3) * (5 - 7)

(D) The infix expression 2 + 3 * 5 – 7

1998 ~ Theoritical qustion

postfix expression

BODMAS

↓↓

order of

unary

$3 * \log(x+1) - \frac{a}{2}$

$= 3 * \log x1+ - a/2$

$= 3 * x1+\log - a/2$

$= 3x1+\log * - a2/$

$= 3x1+\log * a2/-$

$-5*2$

$5-*2$

$5-2*$

unary

Topic : Question

Q Consider the following expression

25 − (80) = −55

(5 2 * 3 3 2 + * +) − (+ * 5 2 * + 3 4 * 5 2)

What will the result after evaluating the expression Where the sub expression in post fix and prefix form and all numbers are single digit numbers

(A) 250 (B) -55 (C) 80 (D) 1000





Slide

Topic : Question

#	postfix		Value
P	2 3 * 4 2 2 * / + 3 5 + -	A	80
Q	6 2 3 + - 3 8 2 / + * 2 $ 3 +	B	-1
R	5 2 * 3 4 + 5 2 * * +	C	52
S	8 2 3 ^ / 2 3 * + 5 1 * -	D	25
R	5 2 * 3 3 2 + * +	E	2
U	10,5 + 60,6,/ * 8,–	F	142

GATE

HW

Telegram

H.W

Slide

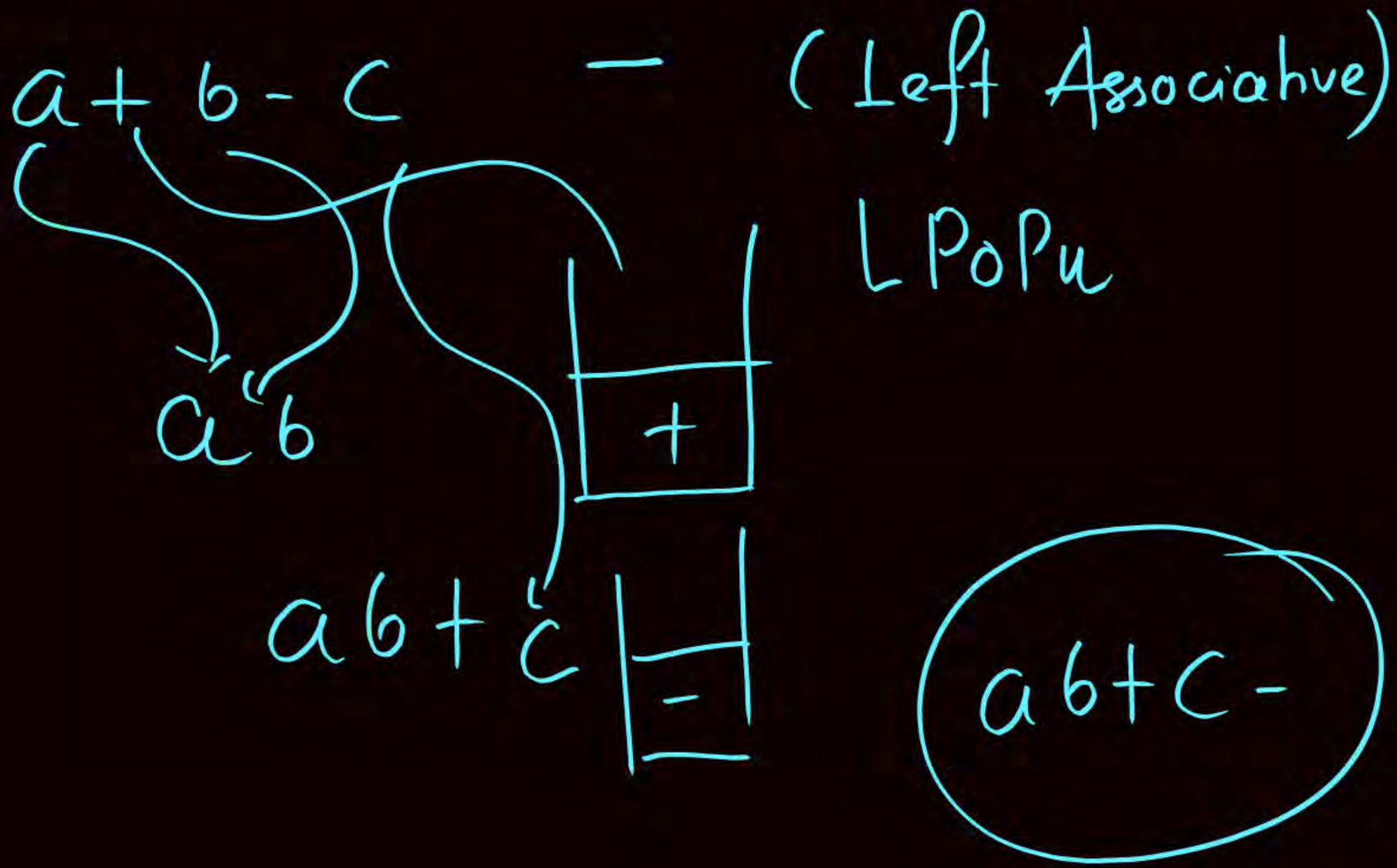


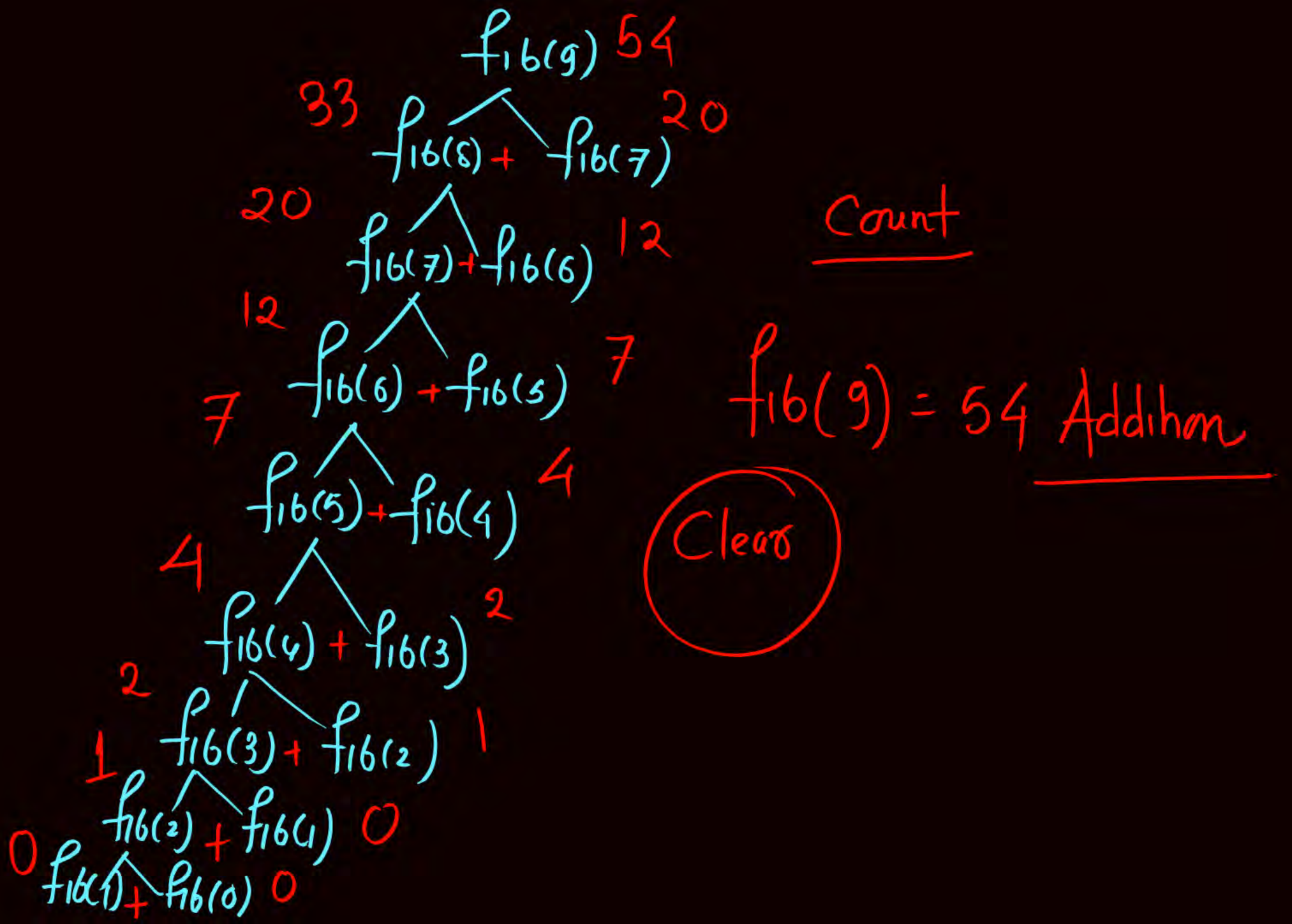
Topic : Question

#	postfix		Value
P	2 3 * 4 2 2 * / + 3 5 + -	A	80
Q	6 2 3 + - 3 8 2 / + * 2 $ 3 +	B	-1
R	5 2 * 3 4 + 5 2 * * +	C	52
S	8 2 3 ^ / 2 3 * + 5 1 * -	D	25
R	5 2 * 3 3 2 + * +	E	2
U	10 5 + 60 6 / * 8 –	F	142

Slide

P-B, Q-C, R-A, S-E, T-D, U-F


a+b-c
– (Left Associative)
LPoPu
a
b
+
ab+c
-
ab+c-


fib(9) 54
33
fib(8) + fib(7)
20
20
fib(7) + fib(6)
12
12
fib(6) + fib(5)
7
7
fib(5) + fib(4)
4
4
fib(4) + fib(3)
2
2
fib(3) + fib(2)
1
1
fib(2) + fib(1)
0
0
fib(1) + fib(0)
0
Count
fib(9) = 54 Addition
Clear



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Evaluation of postfix

Evaluation of prefix

practice problem

Slide

THANK - YOU

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